

P3868R0

Allow `#line` before module declarations

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Abstract

[P1857R3] disallowed any control-lines before the global module fragment. This has turned out to be too strict due to many common tools running the preprocessor separately. We need to allow `#line` directives on any line. This resolves US 55-102.

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§ 1. The Problem

After implementing part of [P1857R3], [Clang soon discovered](#) that it makes the following code ill-formed:

```
#line 1 "A.cppm"  
export module a;
```

This is a problem because tools such as [build2](#), [ccache](#), and [distcc](#), among many others, rely on feeding the preprocessed output back into the compiler. This output generally contains `#line` (or GNU line marker) directives at the start of the file.

§ 2. The Fix

The standard should allow `#line` directives to appear on any line.

§ 3. Wording

Modify [cpp.pre]:

module-file:

group_{opt} pp-global-module-fragment_{opt} pp-module group_{opt} pp-private-module-fragment_{opt}

4 At the start of phase 4 of translation, the *group* of a *pp-global-module-fragment* shall contain neither a *text-line* nor a *pp-import*.

5 At the start of phase 4 of translation, the leading group of a *module-file*, if present, shall consist only of `#line` directives.

§ References

§ Informative References

[P1857R3]

Michael Spencer. *Modules Dependency Discovery*. 11 February 2020. URL:
<https://wg21.link/p1857r3>